CO2-AT1-SCENARIO BASED PROBLEMS

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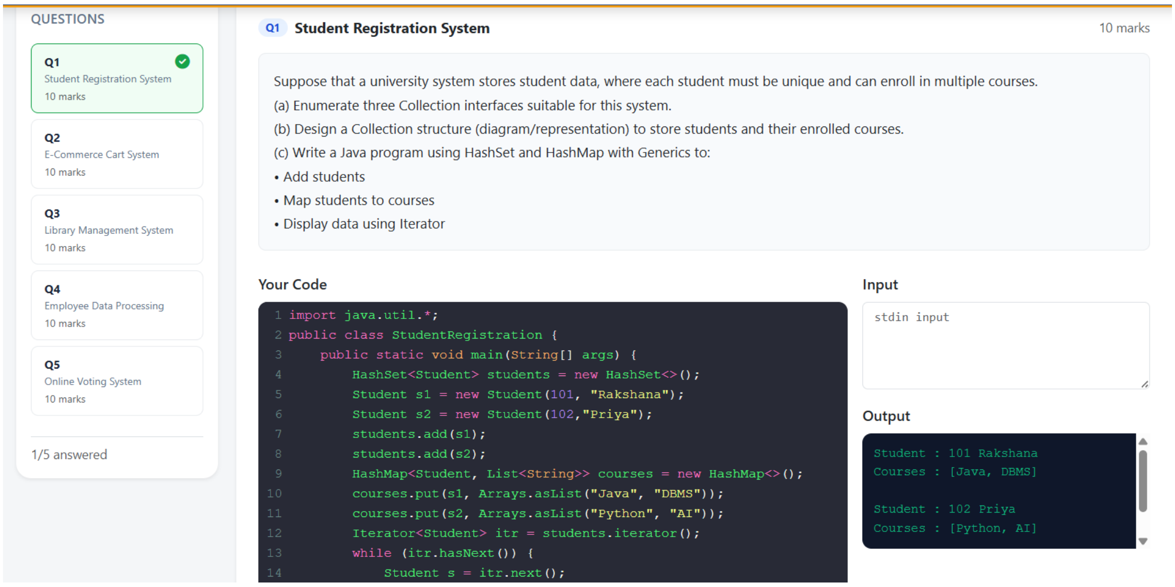
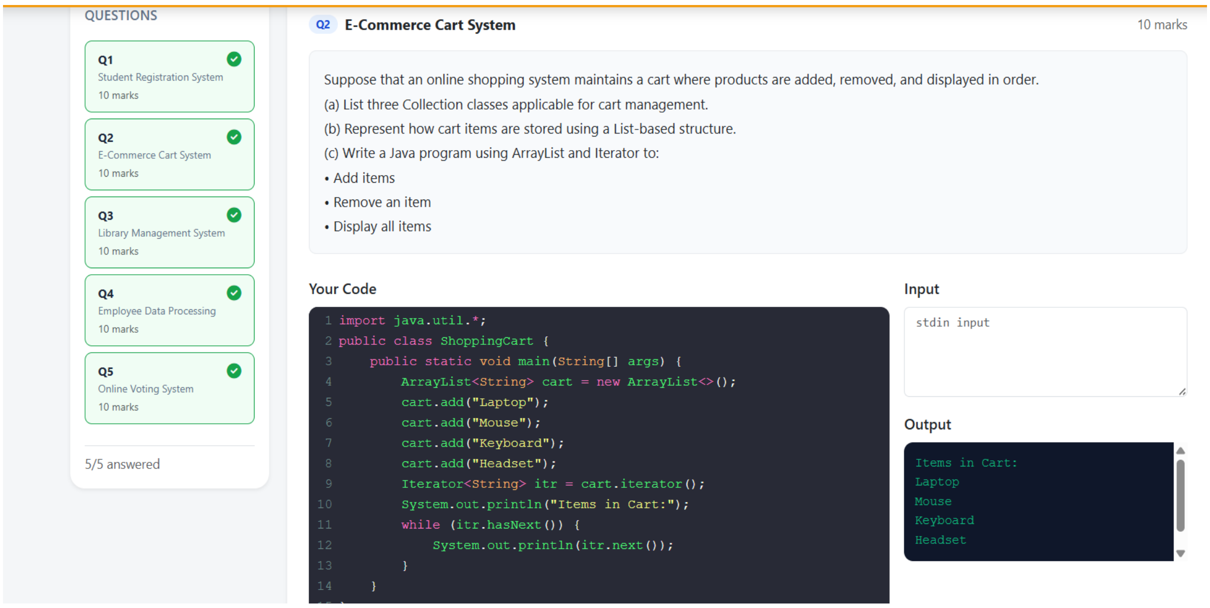
QUESTION-1:



QUESTION-2:



QUESTION-3:

Q3 **Library Management System** 10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.
(a) Name three Map implementations in Java.
(b) Design a schema-like representation using Map for storing ISBN and book details.
(c) Write a Java program using HashMap to:
• Insert book details
• Retrieve a book
• Display all books

Your Code

```
import java.util.*;

public class LibraryManagement {

    public static void main(String[] args) {

        HashMap<String,String> books = new HashMap<>();

        books.put("97812345","Java Programming");
        books.put("97867890","Python Basics");
        books.put("97854321","Data Structures");

        System.out.println("Retrieved Book:");
        System.out.println(books.get("97867890"));
```

Input

```
stdin input
```

Output

```
Retrieved Book:
Python Basics

All Books:
97854321 -> Data Structures
97812345 -> Java Programming
```

QUESTION-4:

Q4 **Employee Data Processing** 10 marks

Suppose that a company processes employee records and filters employees based on salary.
(a) List three advantages of Generics in Collections.
(b) Represent how employee data is stored using List with Generics.
(c) Write a Java program using Iterator to:
• Store employee objects
• Display employees with salary > 50,000

Your Code

```
import java.util.*;
public class EmployeeProcessing{
    public static void main(String[] args){
        ArrayList<Employee> list=new ArrayList<>();
        list.add(new Employee(101,"Ravi",45000));
        list.add(new Employee(102,"Priya",65000));
        list.add(new Employee(103,"Rakshana",70000));
        Iterator<Employee> itr=list.iterator();
        System.out.println("Employees Salary > 50000");
        while(itr.hasNext()){
            Employee e=itr.next();
            if(e.salary>50000){
                System.out.println(e);
            }
        }
```

Input

```
stdin input
```

Output

```
Employees Salary > 50000
102 Priya 65000.0
103 Rakshana 70000.0
```

QUESTION-5:

Q5 **Online Voting System** 10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.
(a) Identify three suitable Collection types for this system.
(b) Design a structure using Set and Map for voters and vote counting.
(c) Write a Java program using HashSet and HashMap to:
• Record votes
• Prevent duplicate voting • Display results

Your Code

```java
import java.util.*;
public class OnlineVoting {
    public static void main(String[] args) {

        HashSet<String> voters = new HashSet<>();

        HashMap<String,Integer> votes = new HashMap<>();

        if(voters.add("Voter1"))
            votes.put("Candidate A",
            votes.getOrDefault("Candidate A",0)+1);

        if(voters.add("Voter2"))
            votes.put("Candidate B",
            votes.getOrDefault("Candidate B",0)+1);
```

Input

```
stdin input
```

Output

```
Duplicate Vote Not Allowed

Voting Results
Candidate B = 1
Candidate A = 2
```